Ling 285
Human Language and Technology
Lecture 3

Dynamical Systems 1

Khalil Iskarous

Department of Linguistics

University of Southern California

Today

- Dynamical Systems in behavior (Animal)
- Dynamical Systems in learning (AI)

DS: 1) Elements, 2) Relations, 3) Goal

DS: 1) Elements, 2) Relations, 3) Goal

- Elements:
  - 8 arms
  - Ground
- Relations between individual arms, ground, and fish are planned by a neural system residing in its head and the rest of its body.
- Goal: Keep den

DS: 1) Elements, 2) Relations, 3) Goal

- Elements:
  - Many parts of tongue
  - Lips
  - Velum
  - larynx
  - Palate
  - Pharyngeal Wall
- Relations between all these planned by the brain.
- Goal: Convey a thought

Some Dynamical Systems in you

- All of your daily voluntary actions, as well as involuntary actions of your heart, pancreas, neurons, etc.... They’re all dynamical systems
- Their elements could be as big as organs or parts theorof downright to cells, and chemicals in them... Biology, Chemistry, and Physics are all about dynamical systems.
- So your speech system is a very complex dynamical system.
- Since there are so many dynamical systems, we can use one dynamical system to study another, due to the underlying similarity, despite extreme difference (We’re going up Diotima’s ladder!)

Marine Biology (2023) 170:99
https://doi.org/10.1007/s00227-023-04243-y

ORIGINAL PAPER

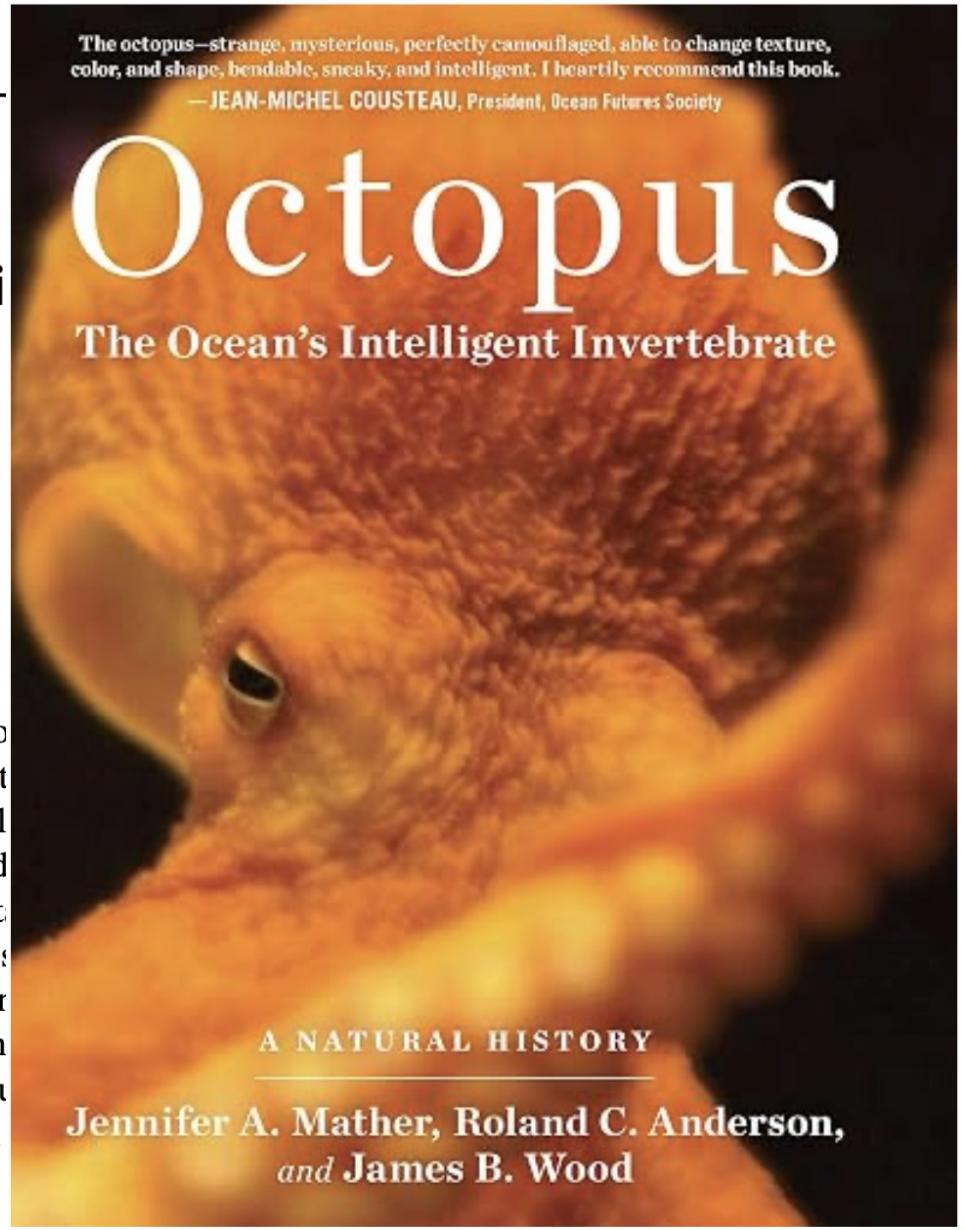
A syntactic analysis of a complex motor acti

Jean Alupay[1] · Jennifer Mather[2] · Khalil Iskarous[3]

Received: 15 February 2023 / Accepted: 1 June 2023
© The Author(s) 2023

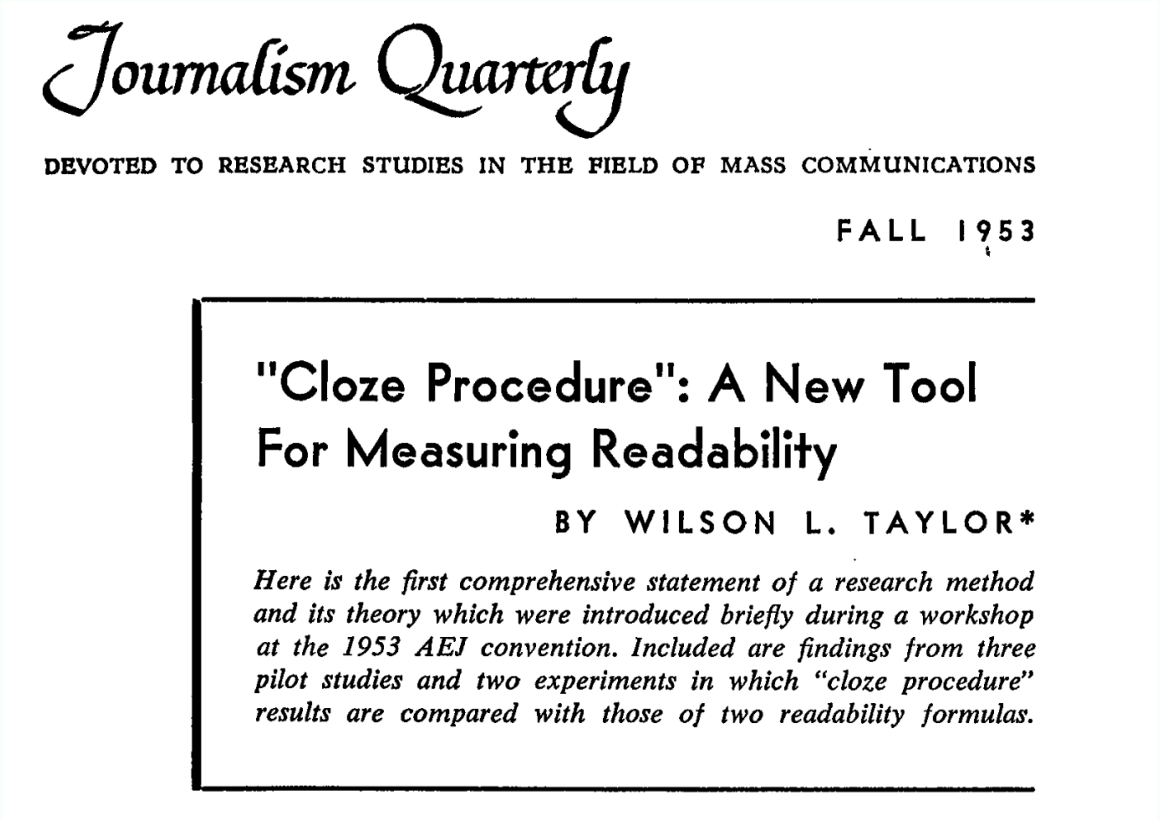
Abstract
The analysis of a natural motor action is always difficult, especially
the same interaction with the environment. We analyzed the behavio
Okinawa, Japan, which used the kinematic primitives of rotation and t
kinematic behaviors serially and in parallel to 'slap' at fish in the wil
arm movement were measured in both external and animal-centered
the fish. By combining these primitives, the octopus is able to maint
freedom, a concept we term 'flexible rigidity'. This slapping action
view of motor behavior, as well as their idea that motor primitives car
octopus's ability to use sensory feedback from the position of a movin
tives, allows for the building of complex actions at dynamic equilibriu
to a more realistic view of how a complex behavior allows an animal

Keywords Octopus · Muscular hydrostat · Motor control · Abdopus



What is AI?

- AI is a model of several animal psychological skills:
  - Knowing
  - Recognizing
  - Attending
  - Acting
  - Learning



Journalism Quarterly

DEVOTED TO RESEARCH STUDIES IN THE FIELD OF MASS COMMUNICATIONS

FALL 1953

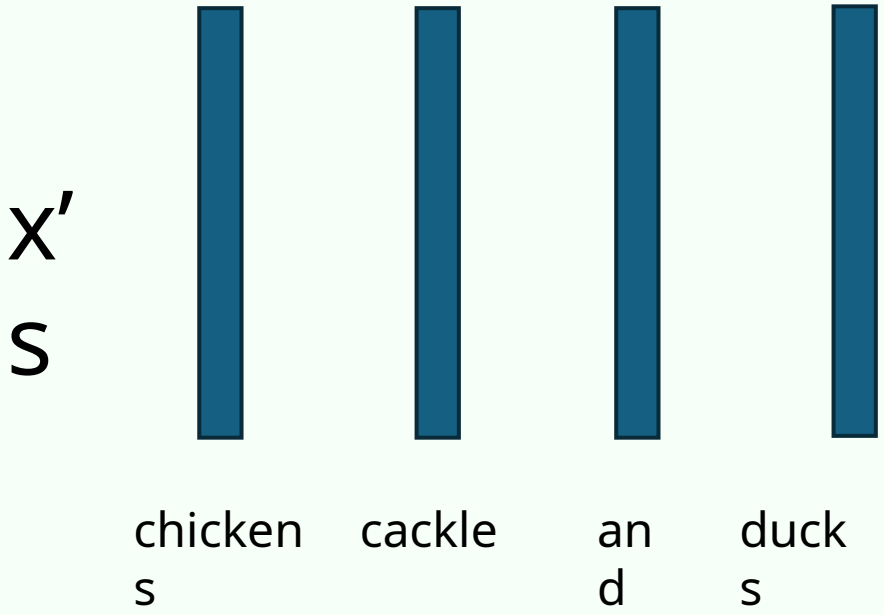
"Cloze Procedure": A New Tool For Measuring Readability

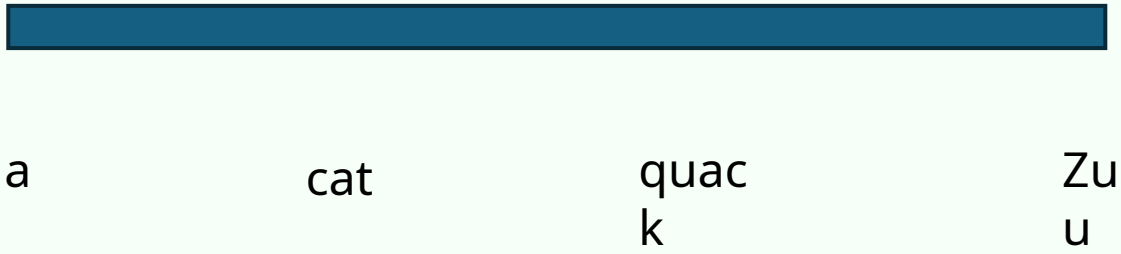
BY WILSON L. TAYLOR*

Here is the first comprehensive statement of a research method and its theory which were introduced briefly during a workshop at the 1953 AEJ convention. Included are findings from three pilot studies and two experiments in which "cloze procedure" results are compared with those of two readability formulas.

GPT

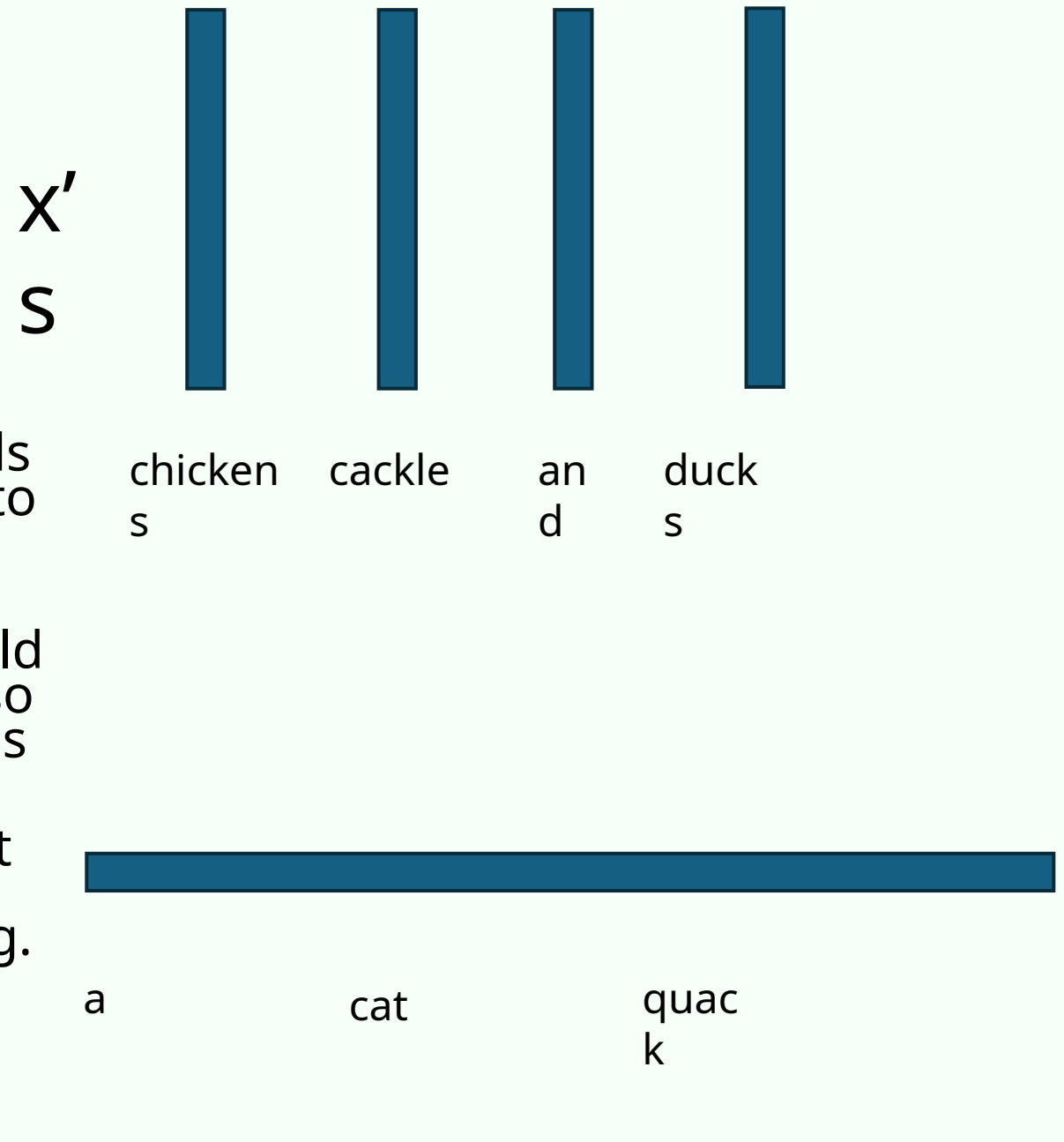
- Each word is a *vector* with thousands of numbers.
- We will call them x's or *tokens.*
- AI **knowledge** about speech/language whatever is in a few trillion number called *weights* or w's
- AI then uses mostly +, - , × , ÷ to combine its knowledge W's with the vectors you give it
- The calculations result in the system predicting a probability distribution function that say: this is my guess for the next word!
- How the W's combine with the X's is a dynamical system that we will meet later
- We will now go to an easier dynamical system: Learning!





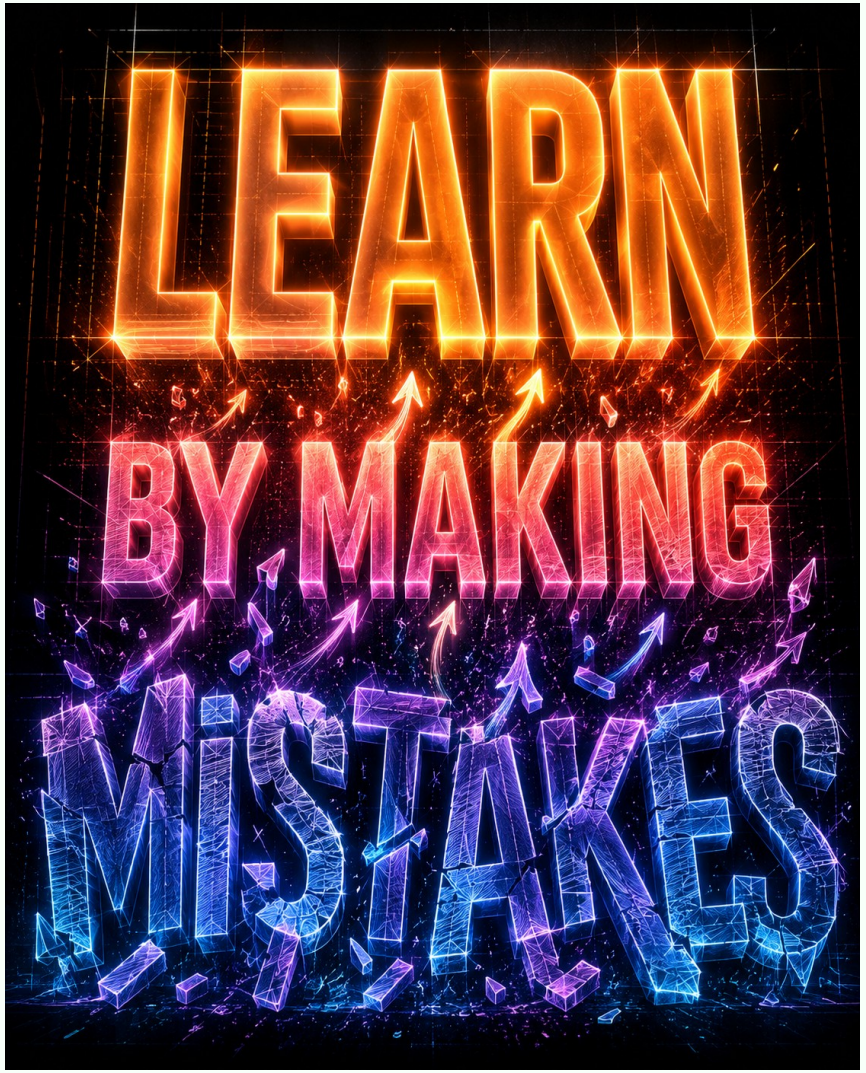
Where do W's (knowledge) come from????

- Fake it till you make it!
- Start out with random, yes totally random, completely random W's-- any number you want!
- We give the system a sequence of words "chickens cackle and ducks " and ask it to predict the next word by combining its W's with the input vectors
- Since we know that the next word should be "quack", we tell it: Change your W's so that your output pdf looks more like this
- Keep exposing the system to more sequences and ask it to predict the next word, billions of times, and keep changing its W's to improve its guessing.
- That's how LLMs *learn* and *know*!



2 Tricks to AI Learning





ChatGP

What's the big deal? Averaging is so easy

- What's the average of 10,12 and 14? $\frac{10+12+14}{3} = 12$
- But this won't work with more complex things like speech, language, biological molecules, scenes, etc., the things AI does…
- The averaging method we develop will work like this:
  - We have a list of numbers
  - We make some wild guess x about their average
  - We seek to improve x by exposure to the data. How?
    - We take one of the numbers in the list and examine it with respect to the current x
    - We now improve x by making it a bit closer to the number from the data we observed. This is observation-based error learning
    - Repeat with another number!

SGD starts with a wild guess

What's the average—*the essence*—of:

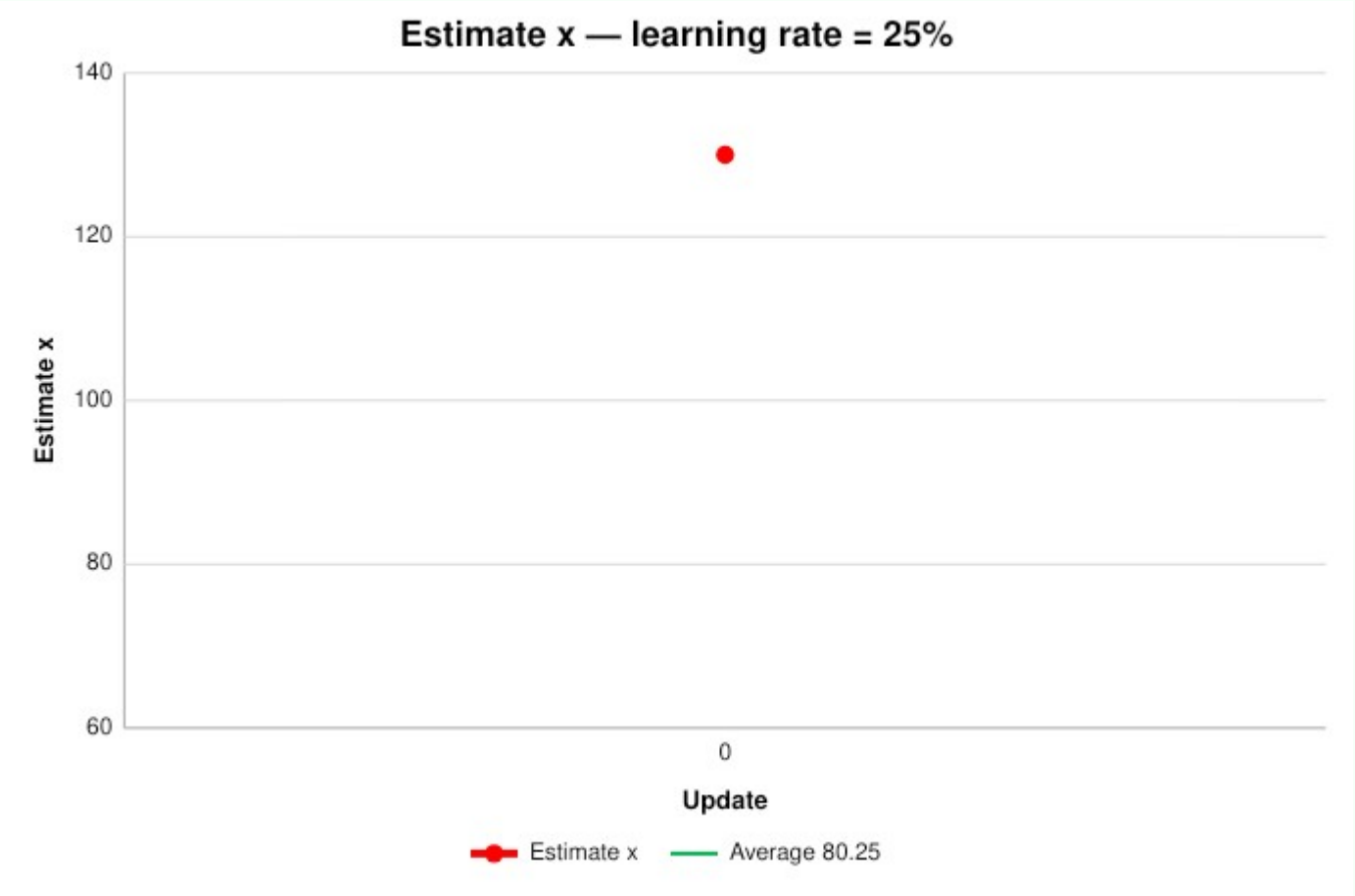
[78, 67, 98, 85, 76, 68, 98, 72]

Pick any random number x as the estimate.

At time 0:

x[0] = 130

Learning rate γ = 25%



One observation creates error

Randomly pick one number d from the data.

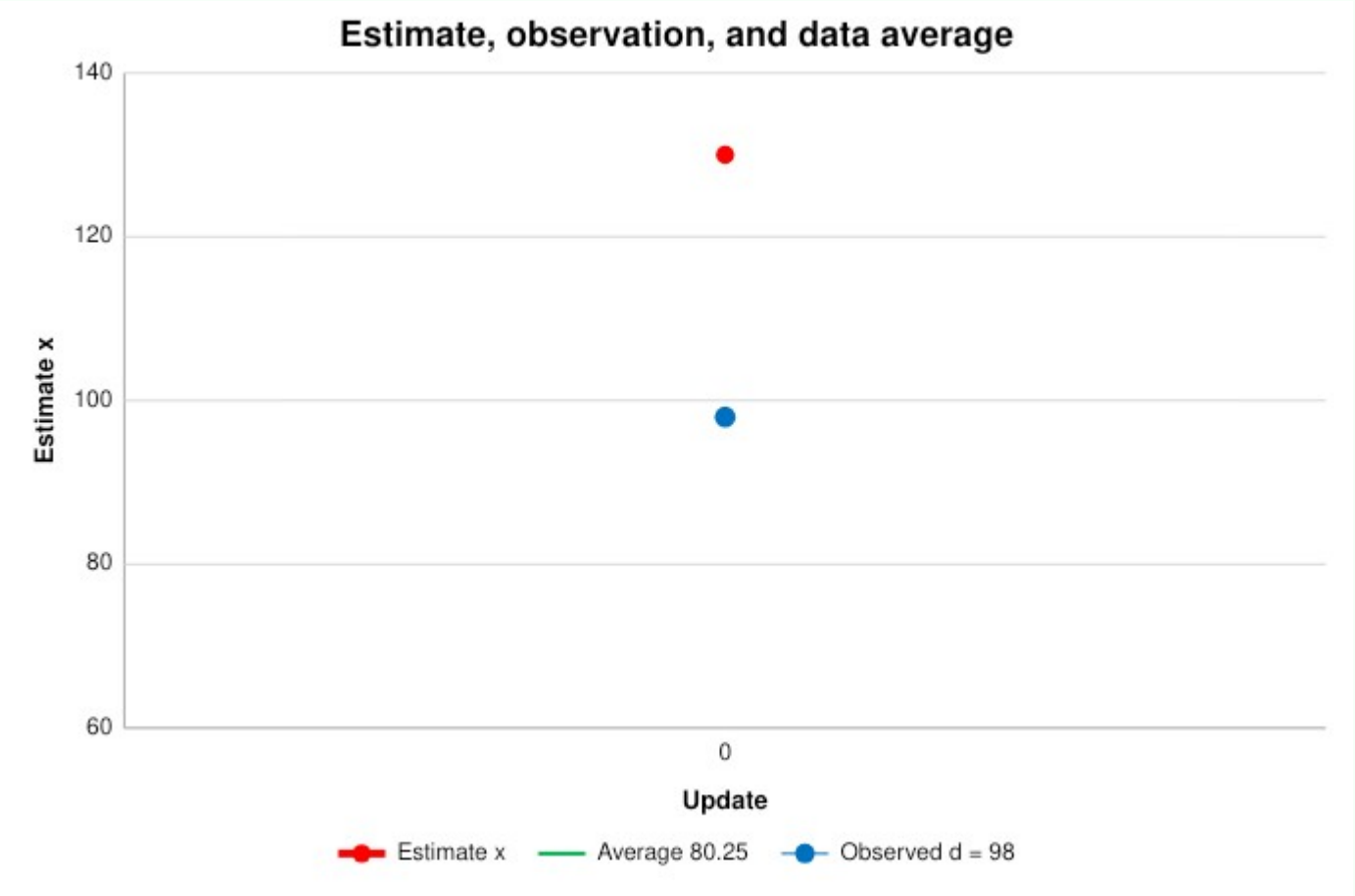
d[0] = 98

Current estimate: x[0] = 130

Discrepancy / Loss:

130 − 98 = 32

The average should be close to the data.



Move 25% of the error

Random observation: d[0] = 98

Discrepancy / Loss:

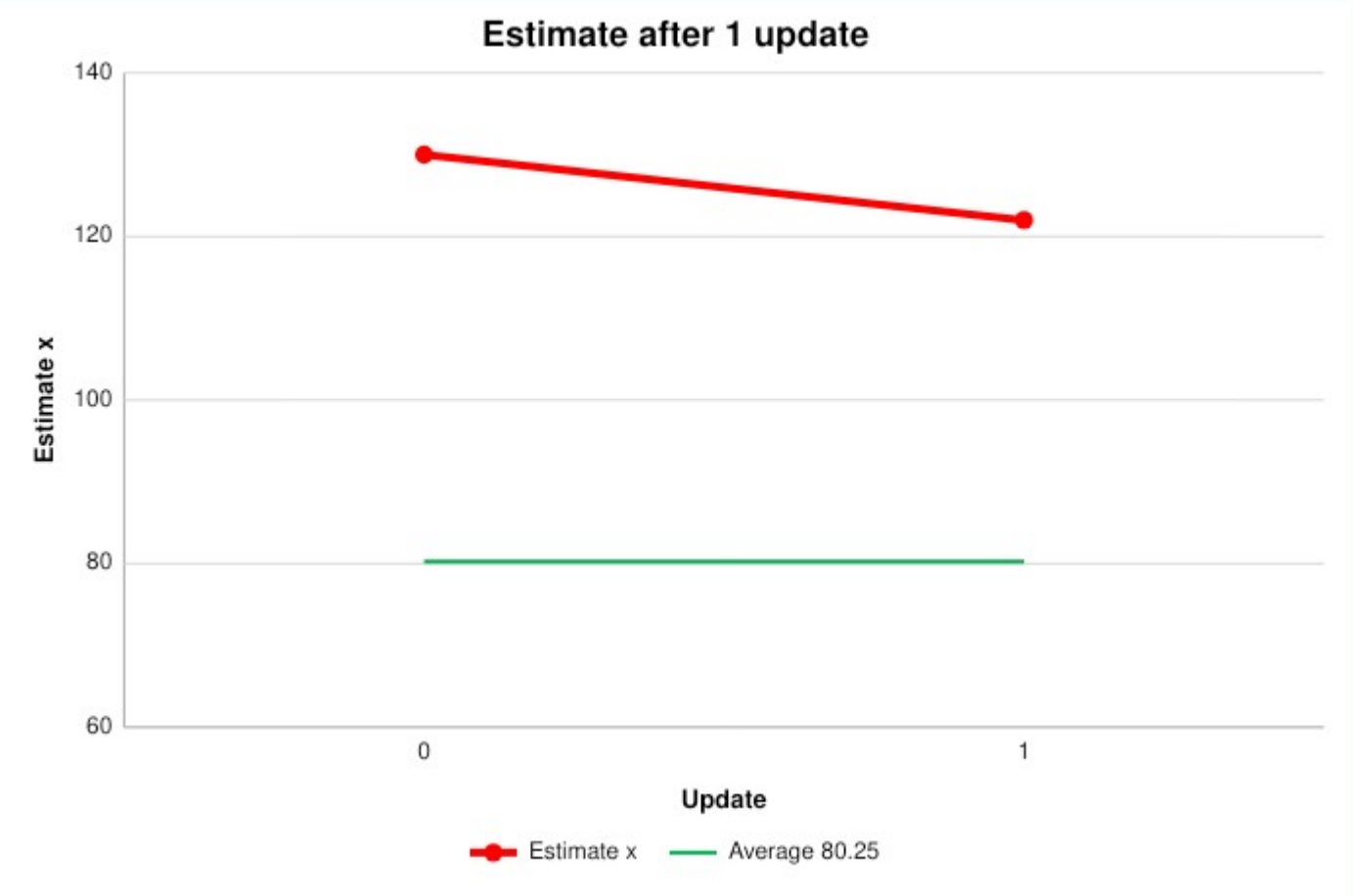
130 − 98 = 32

Subtract 25% of that discrepancy:

x[1] = 130 − 0.25(32)

= 122

The estimate moves toward this observation.



Repeat the correction

Random observation: d[1] = 85

Discrepancy / Loss:

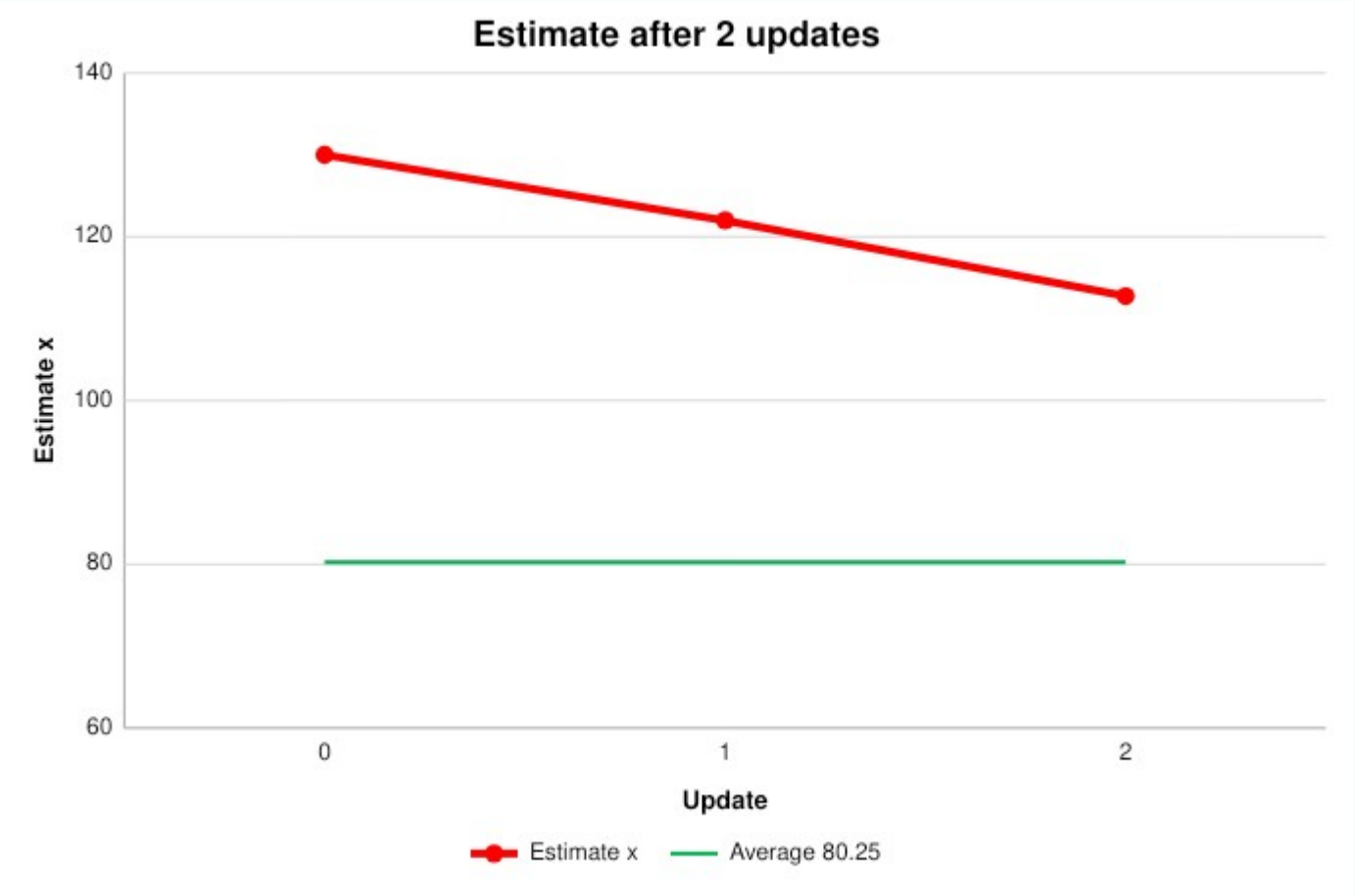
122 − 85 = 37

Subtract 25% of that discrepancy:

x[2] = 122 − 0.25(37)

= 112.75

The estimate moves toward this observation.



Repeat the correction

Random observation: d[2] = 76

Discrepancy / Loss:

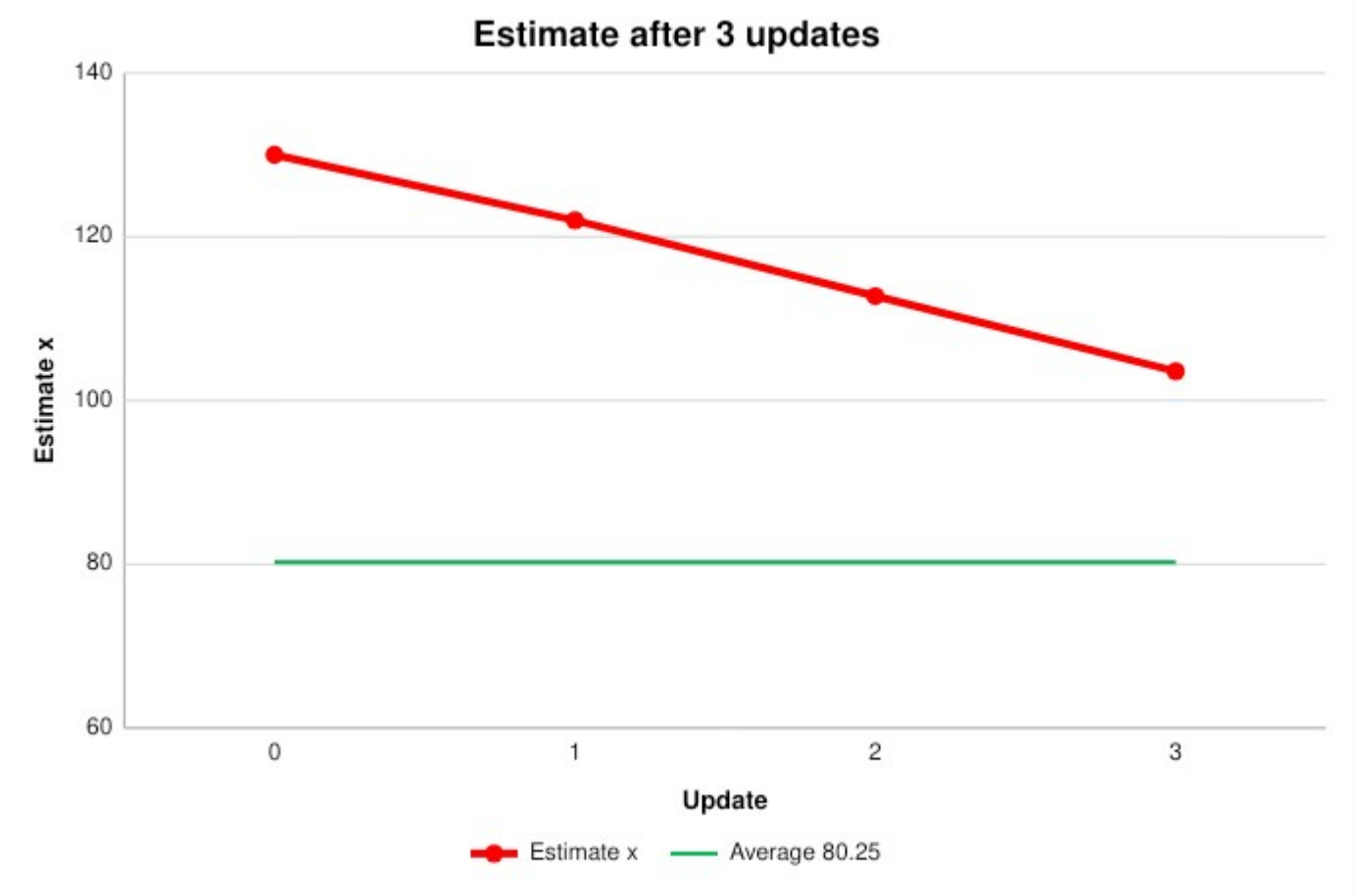
112.75 − 76 = 36.75

Subtract 25% of that discrepancy:

x[3] = 112.75 − 0.25(36.75)

= 103.563

The estimate moves toward this observation.



Repeat the correction

Random observation: d[3] = 68

Discrepancy / Loss:

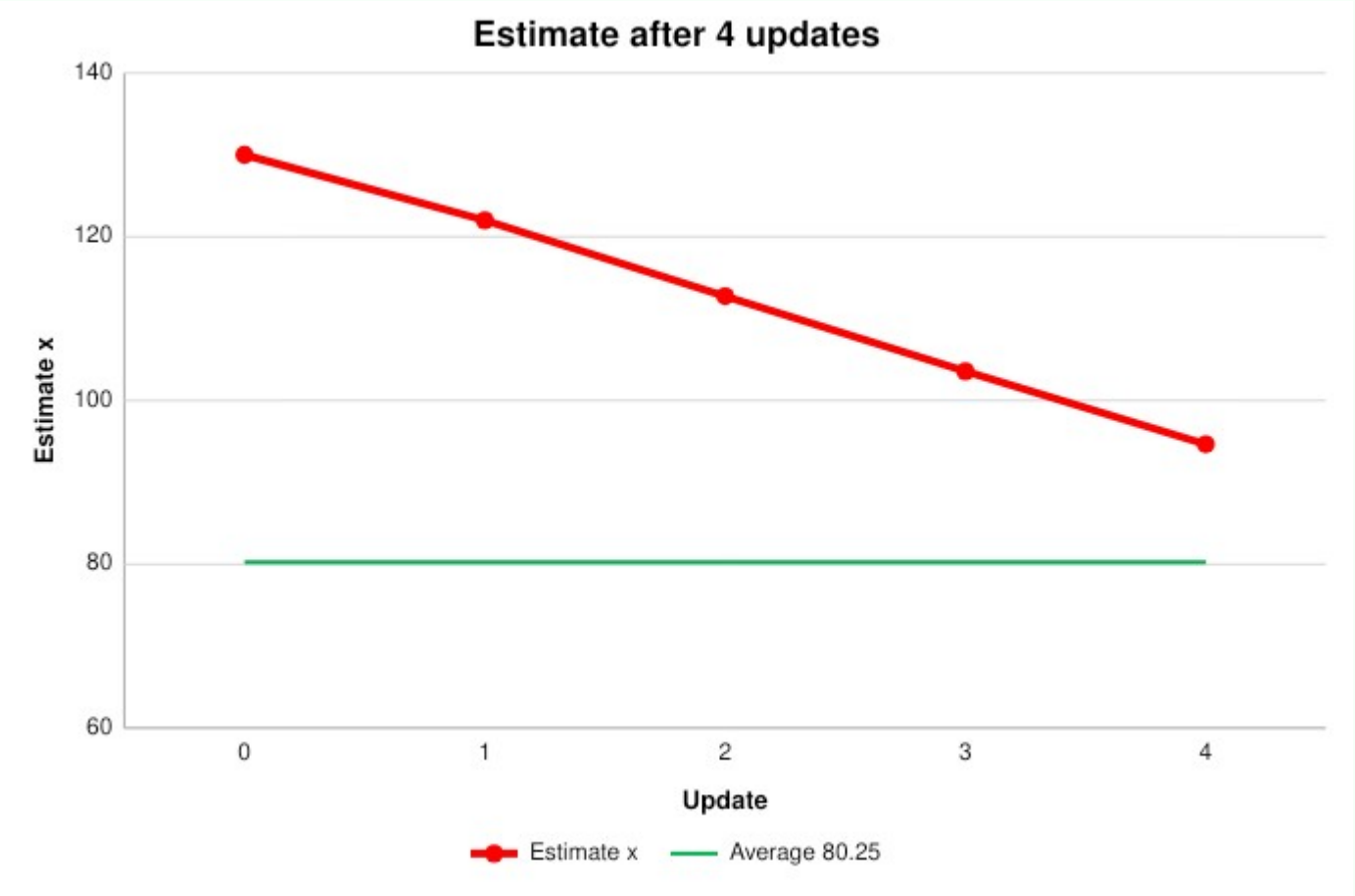
103.563 − 68 = 35.563

Subtract 25% of that discrepancy:

x[4] = 103.563 − 0.25(35.563)

= 94.672

The estimate moves toward this observation.



A high datum pulls x upward

Random observation: d[4] = 98

Discrepancy / Loss:

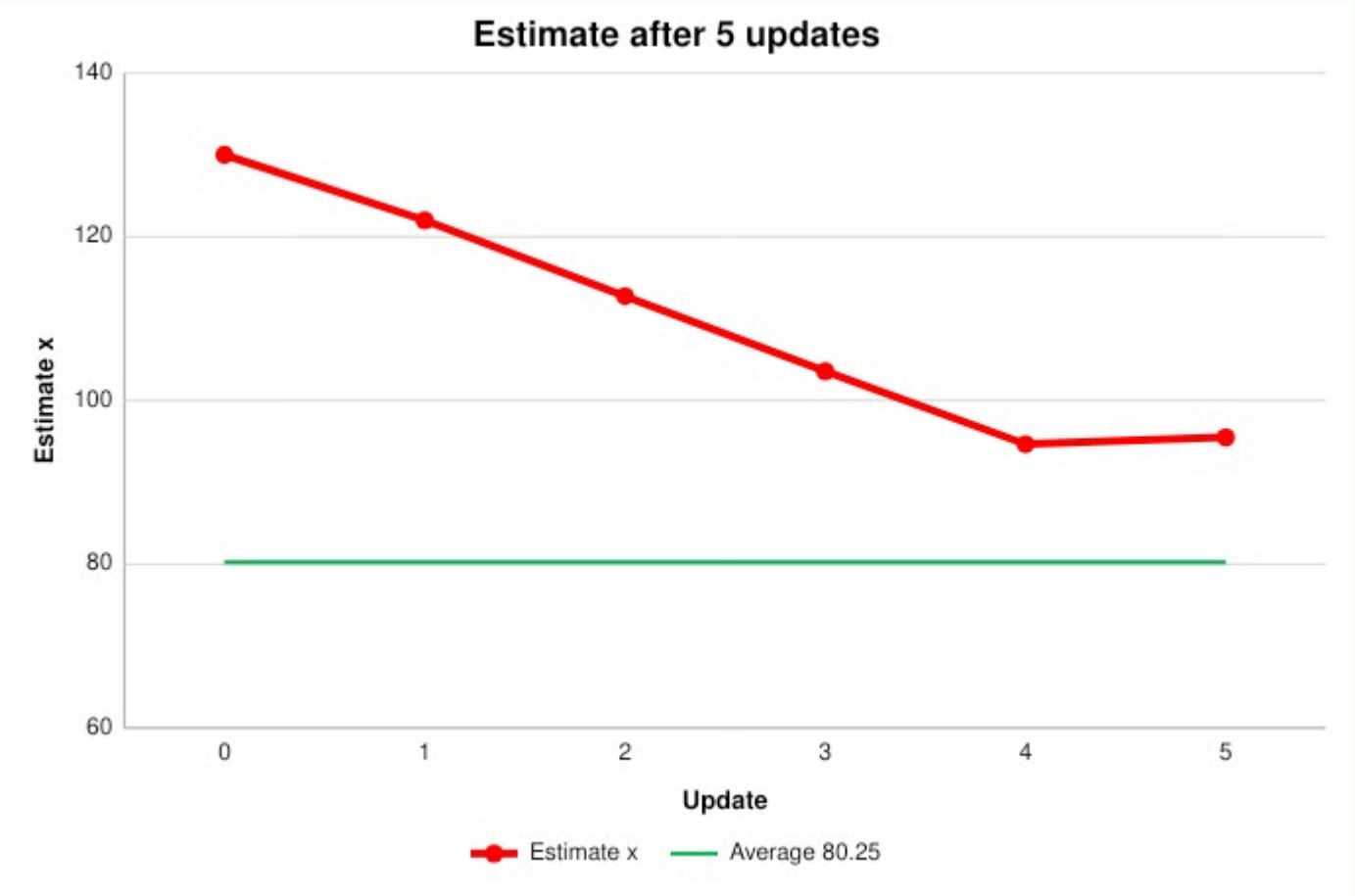
94.672 − 98 = -3.328

Subtract 25% of that discrepancy:

x[5] = 94.672 − 0.25(-3.328)

= 95.504

A negative discrepancy makes subtraction move x upward.



Repeat the correction

Random observation: d[5] = 72

Discrepancy / Loss:

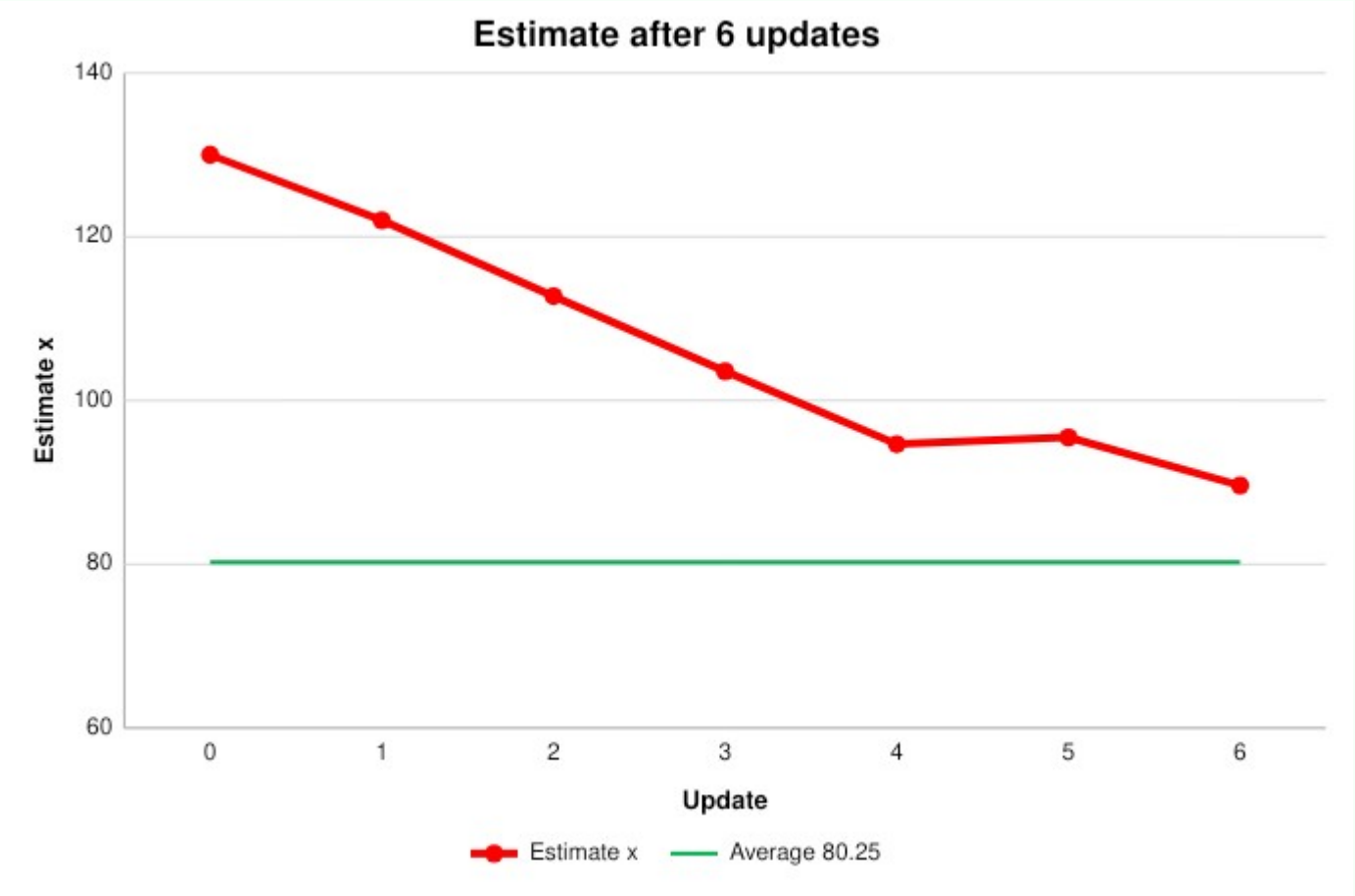
95.504 − 72 = 23.504

Subtract 25% of that discrepancy:

x[6] = 95.504 − 0.25(23.504)

= 89.628

The estimate moves toward this observation.



Repeat the correction

Random observation: d[6] = 78

Discrepancy / Loss:

89.628 − 78 = 11.628

Subtract 25% of that discrepancy:

x[7] = 89.628 − 0.25(11.628)

= 86.721

The estimate moves toward this observation.



Repeat the correction

Random observation: d[7] = 67

Discrepancy / Loss:

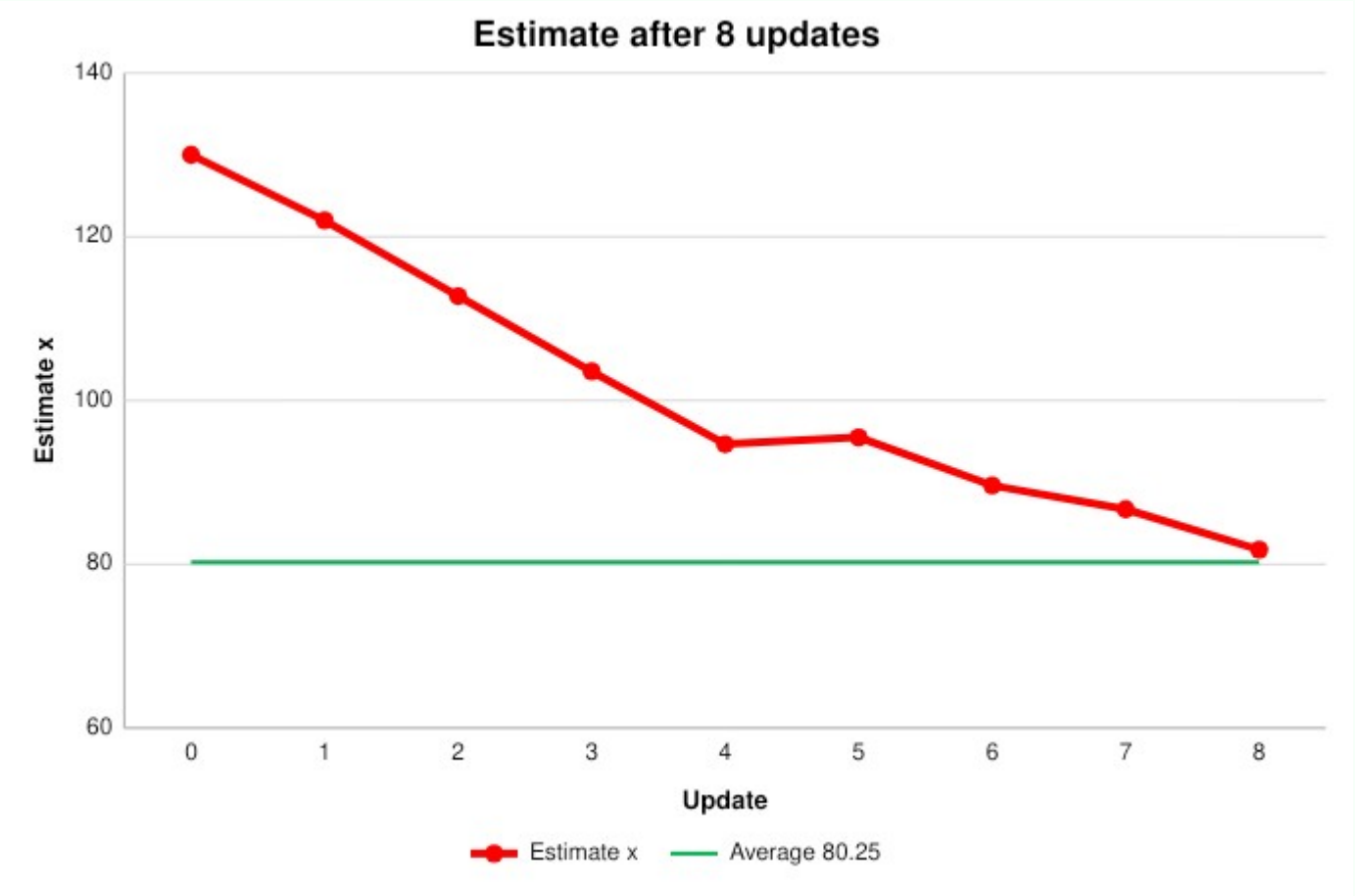
86.721 − 67 = 19.721

Subtract 25% of that discrepancy:

x[8] = 86.721 − 0.25(19.721)

= 81.791

The estimate moves toward this observation.



A high datum pulls x upward

Random observation: d[8] = 98

Discrepancy / Loss:

81.791 − 98 = -16.209

Subtract 25% of that discrepancy:

x[9] = 81.791 − 0.25(-16.209)

= 85.843

A negative discrepancy makes subtraction move x upward.



Repeat the correction

Random observation: d[9] = 85

Discrepancy / Loss:

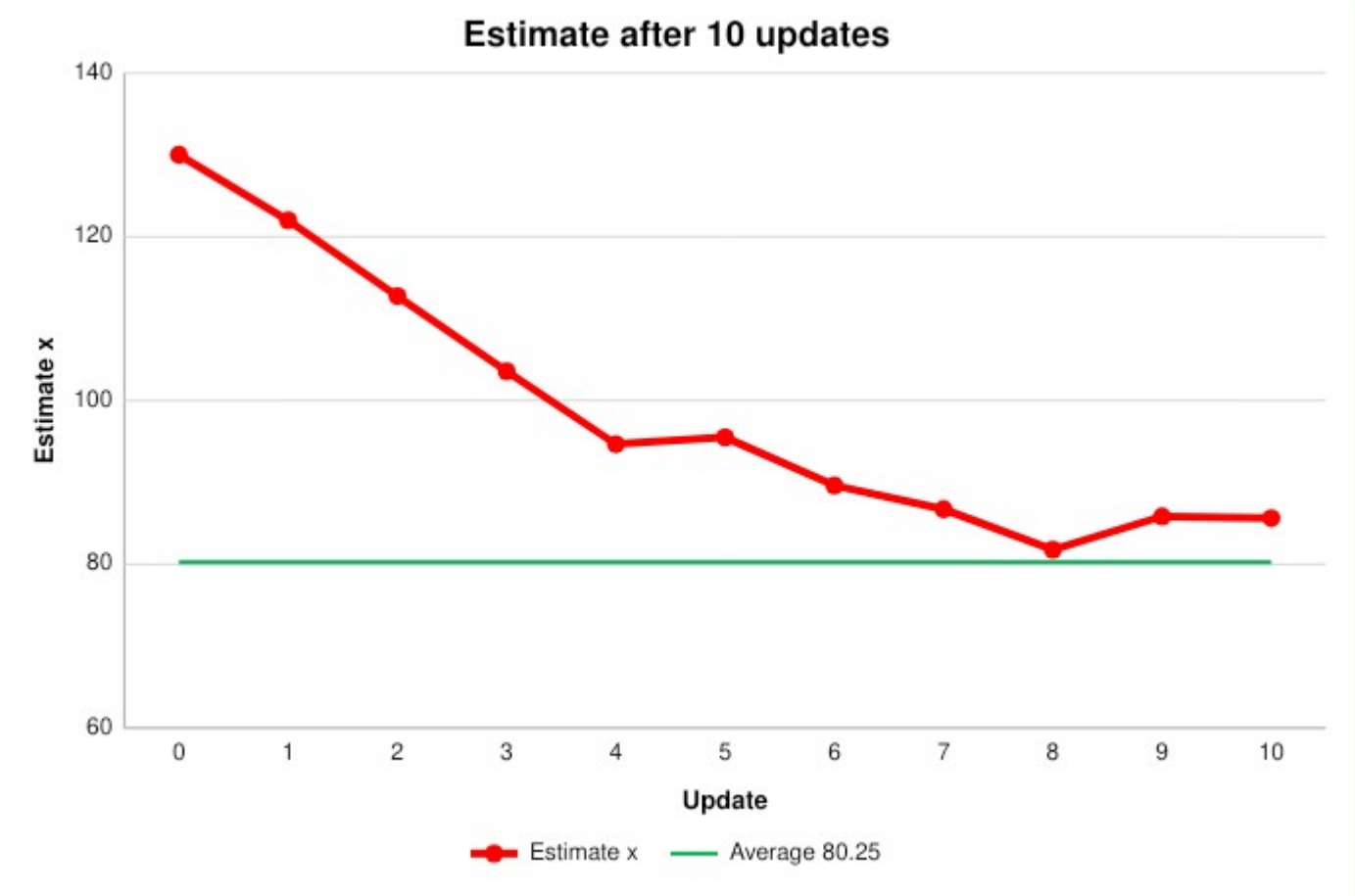
85.843 − 85 = 0.843

Subtract 25% of that discrepancy:

x[10] = 85.843 − 0.25(0.843)

= 85.632

The estimate moves toward this observation.



Repeat the correction

Random observation: d[10] = 76

Discrepancy / Loss:

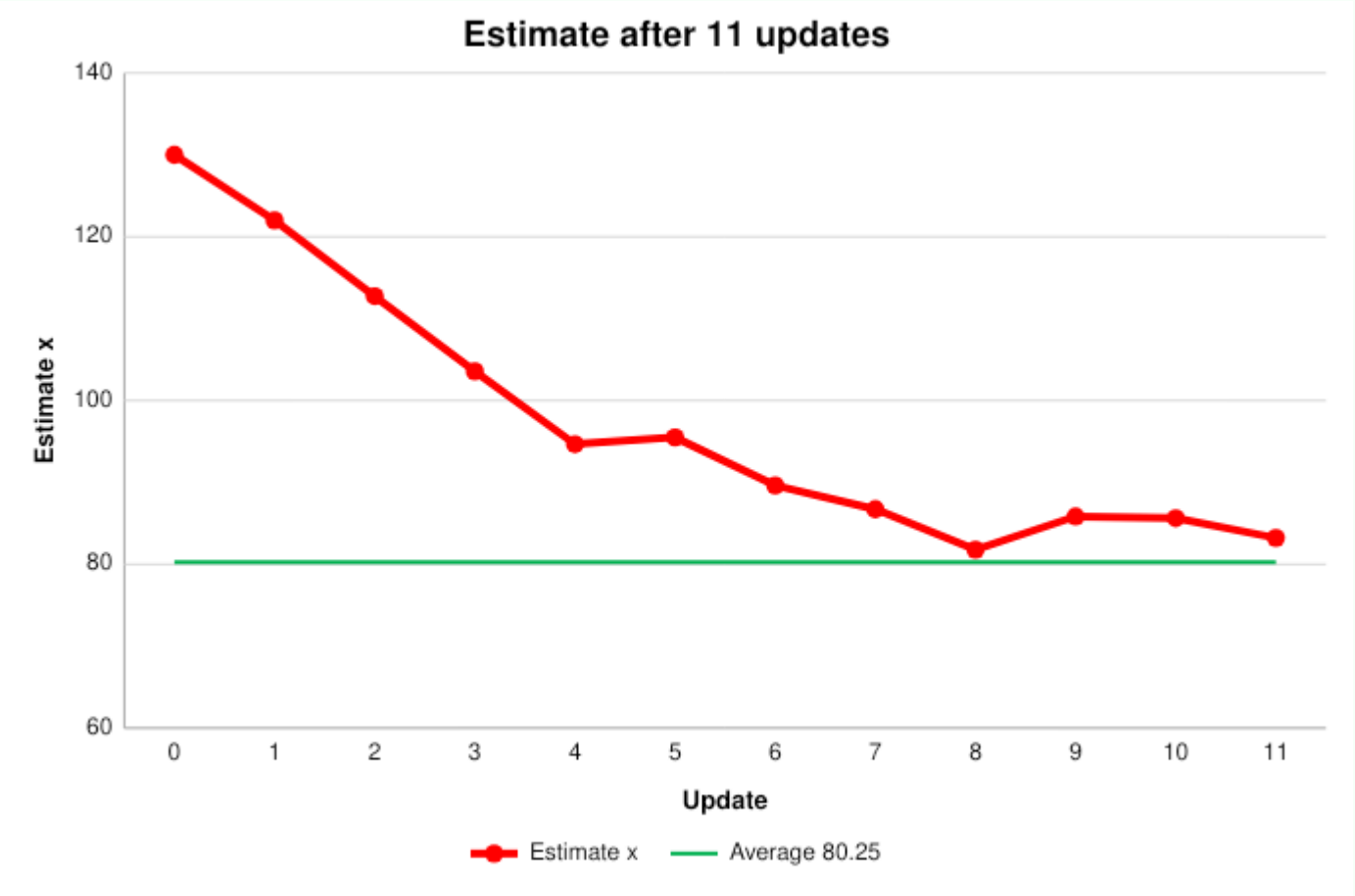
85.632 − 76 = 9.632

Subtract 25% of that discrepancy:

x[11] = 85.632 − 0.25(9.632)

= 83.224

The estimate moves toward this observation.



Repeat the correction

Random observation: d[11] = 68

Discrepancy / Loss:

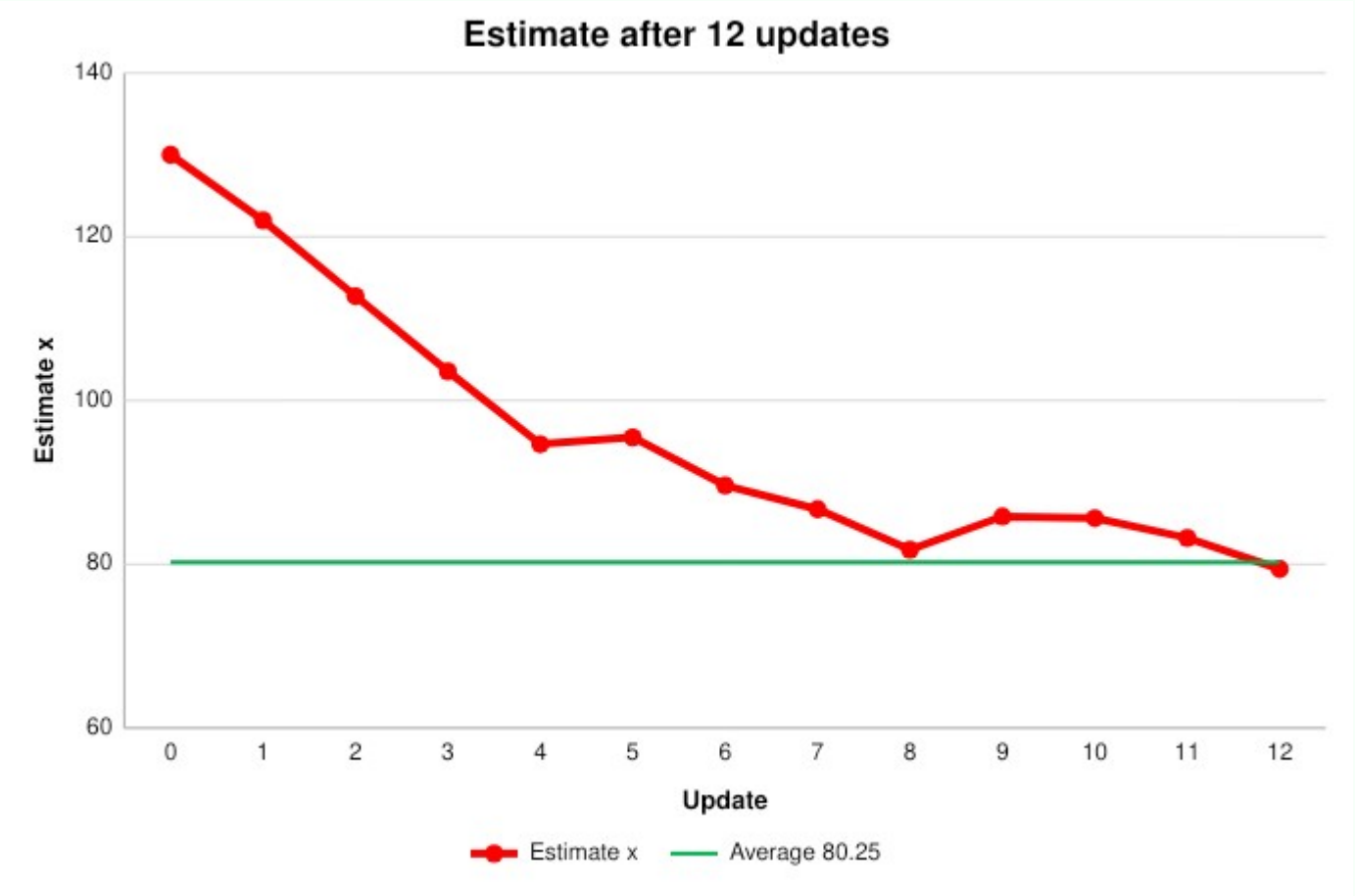
83.224 − 68 = 15.224

Subtract 25% of that discrepancy:

x[12] = 83.224 − 0.25(15.224)

= 79.418

The estimate moves toward this observation.



That is SGD

1) Make an initial estimate / guess.

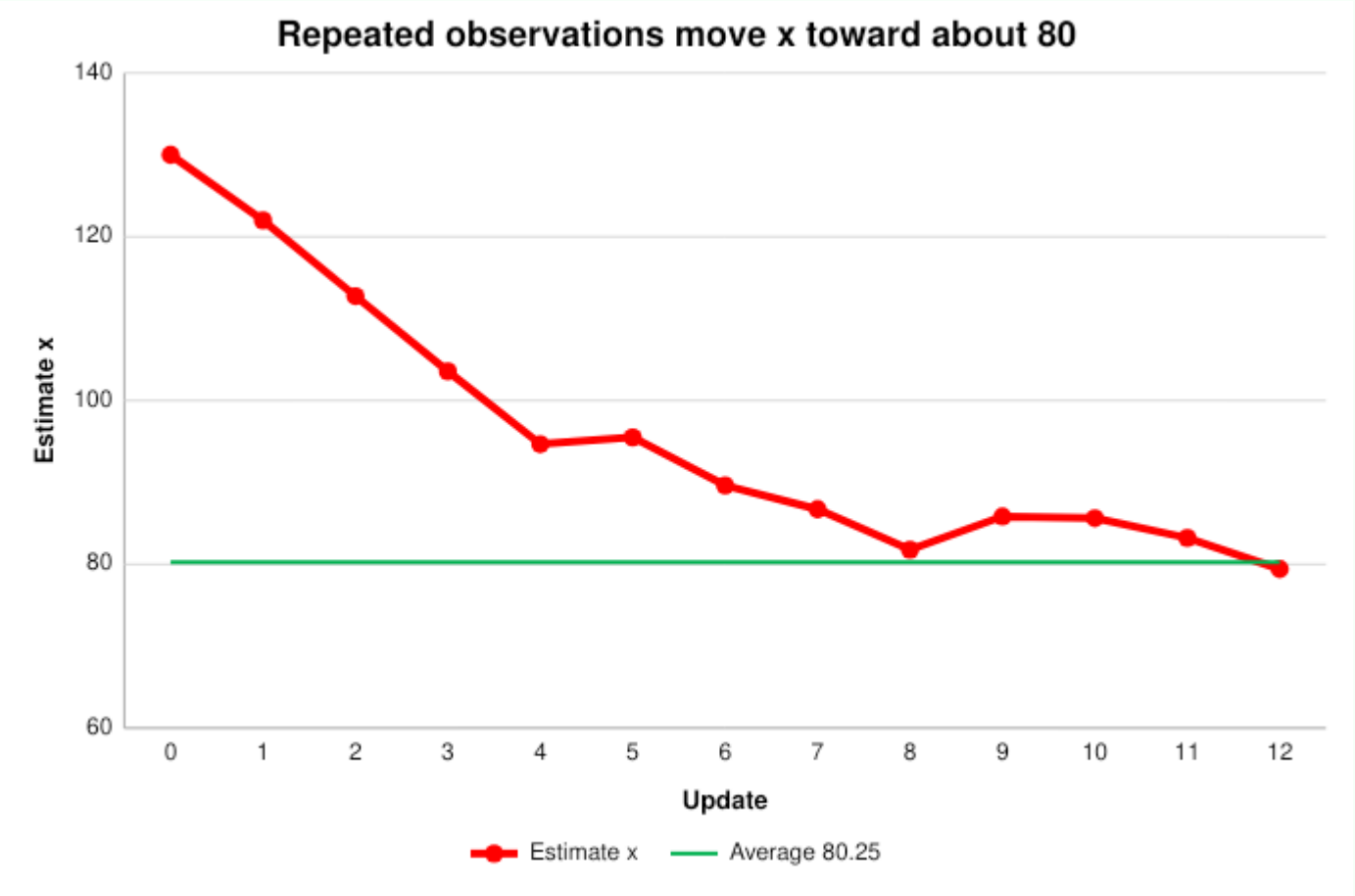
2) Use data to improve the estimate.

Stochastic = random observation

Gradient = Loss-related direction

Descent = reduce the mistake

The path jitters; its tendency is toward the average.



Estimate ↓ means Loss ↓

Second data set:

[62, 70, 78, 82, 90, 98]

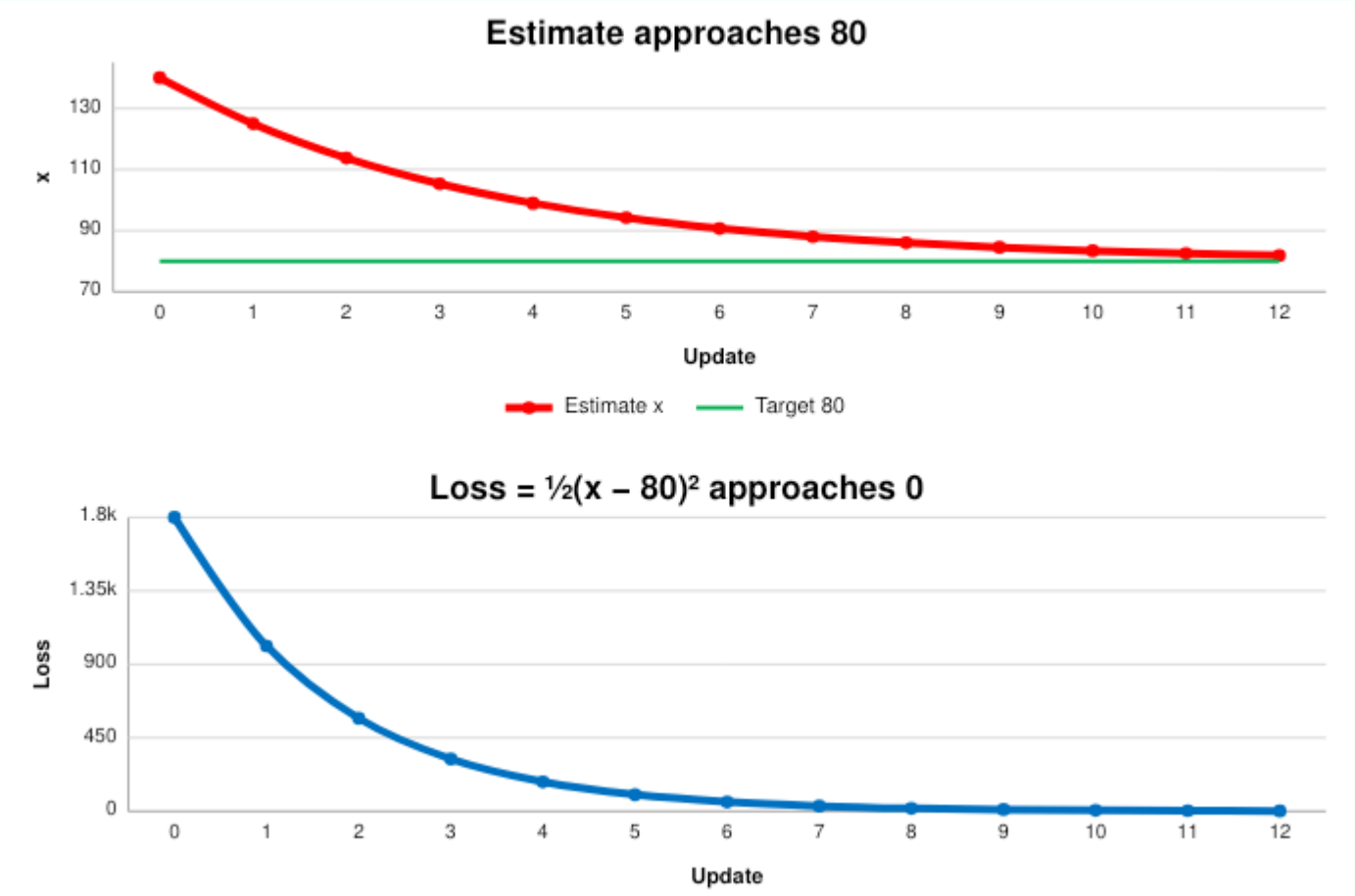
Average = 80

The mean motion of SGD is:

$x[t+1] = x[t] - 0.25(x[t] - 80)$

Loss: $L[t] = \frac{1}{2}(x[t] - 80)^2$

Start: x[0] = 140



Estimate ↑ while Loss ↓

Same data. Same γ = 25%.

Only the initial condition changes:

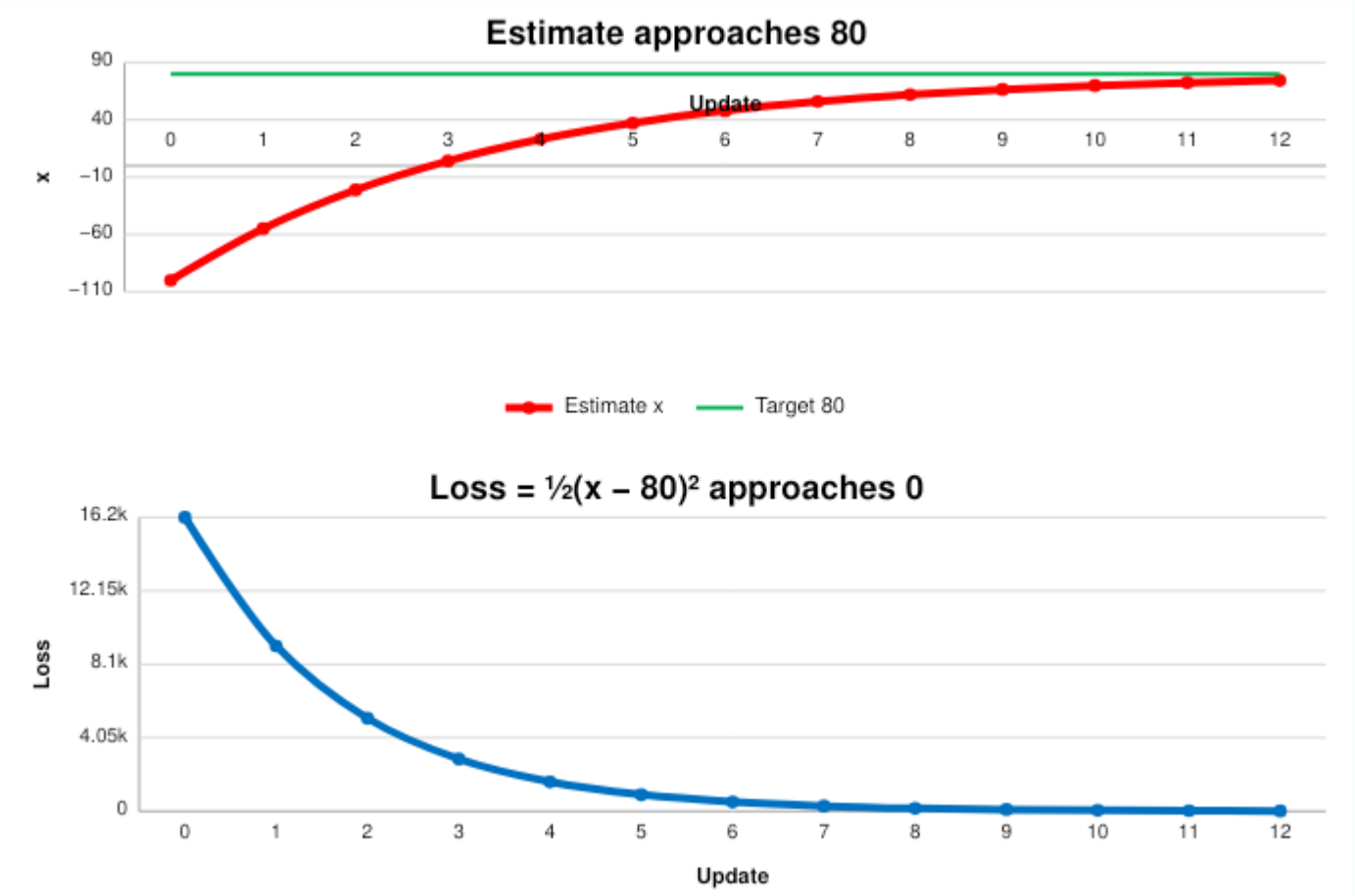
x[0] = −100

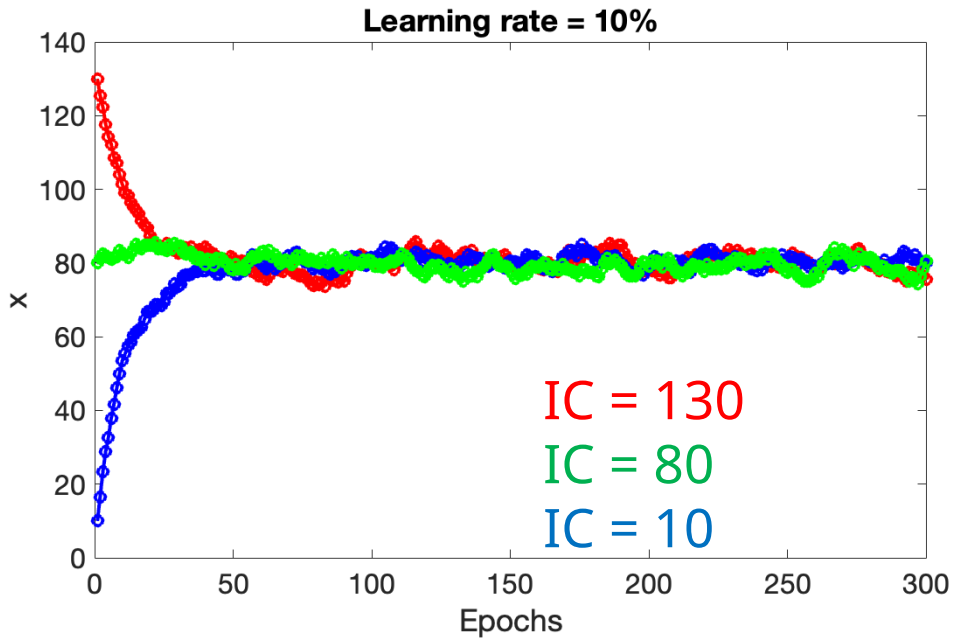
The estimate rises toward 80.

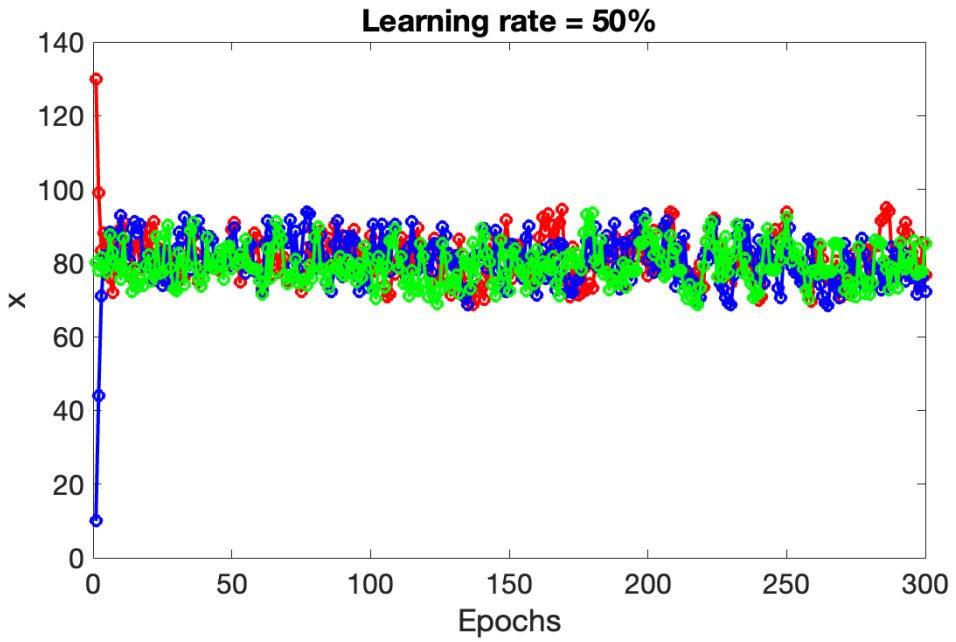
The discrepancy shrinks.

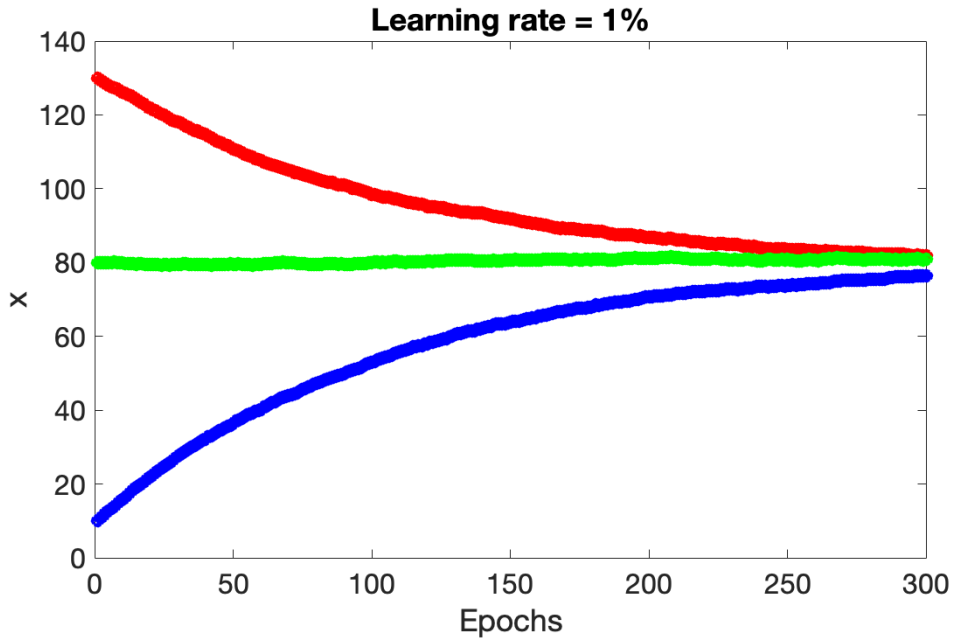
So ½ discrepancy² falls toward 0.

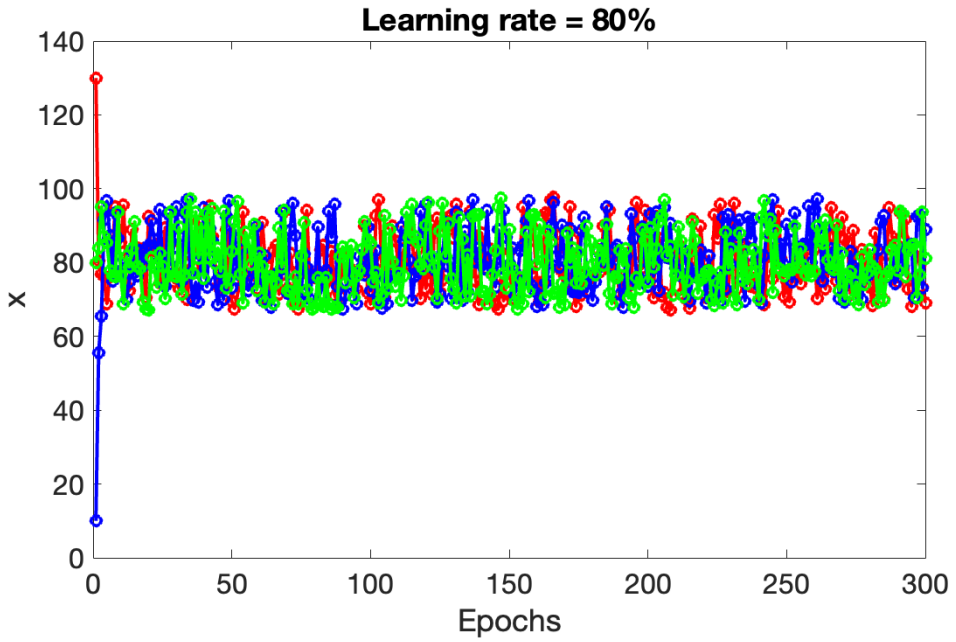
Direction changes; learning does not.




Learning rate = 10%
x
Epochs
IC = 130
IC = 80
IC = 10


Learning rate = 50%
x
Epochs


Learning rate = 1%
x
Epochs


Learning rate = 80%
x
Epochs

Fundamental Learning Tradeoff

- $\gamma \uparrow$ Speed ↑ Accuracy ↓
- $\gamma \downarrow$ Speed ↓ Accuracy ↑
- ☹️
- Thousands of people for generations make their living by finding better way to *optimize* so that
  - Speed ↑
  - Accuracy ↑

